Li Lai

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CV updated: 2026-6-7

Work Experience

2025–Present **Xiamen University**, School of Mathematical Sciences.
Associate Professor

2023–2025 **Peking University**, Beijing International Center for Mathematical Research.
Postdoc, Mentor: Liang Xiao

2020–2021 **Fudan University**, School of Mathematical Sciences.
Research Assistant, Mentor: Yijun Yao

Education

2021–2023 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
Ph.D. Mathematics, Advisor: Pin Yu

2014–2020 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
M.S. Mathematics, Advisor: Pin Yu

2010–2014 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
B.S. Mathematics

Research Interests

I mainly work in the area of transcendental number theory and Diophantine approximation. I am particularly interested in odd zeta values $\zeta(2k+1)$, p-adic odd zeta values $\zeta_p(2k+1)$, and multiple zeta values $\zeta(k_1, k_2, \ldots, k_r)$.

Selected Publications and Preprints

4. Li Lai and Johannes Sprang,
On the $P(t)$-adic Littlewood conjecture in odd characteristics,
arXiv:2606.00633

3. Li Lai, Johannes Sprang and Wadim Zudilin,
A note on the irrationality of $\zeta_2(5)$,
arXiv:2505.05005v2

2. Li Lai, Jiong-Yue Li and Pin Yu,
On the rigidity of stationary charged black holes: small perturbations of the non-extremal Kerr-Newman family,
J. Differential Geom. 125 (2023), no. 3, 553–612.
arXiv:1911.10560

1. Li Lai and Pin Yu,
A note on the number of irrational odd zeta values,
Compos. Math. 156 (2020), no. 8, 1699–1717.
arXiv:1911.08458

Awards and Honors

2012 Nianzeng Sun Mathematical Analysis Award (at Tsinghua University)

2010 51^{st} International Mathematical Olympiad: Gold Medal

Teaching

Spring 2026 Exercise Course of Mathematical Analysis, Xiamen University

Fall 2024 Rational Functions and the Irrationality of Odd Zeta Values, Mini Course, Peking University

Spring 2024 Advanced Mathematics B (2), Peking University

Spring 2021 Rational Functions and the Irrationality of Odd Zeta Values, Short Course, Fudan University

Seminar/Conference (Co)Organized

July 27–31, 2026 The 5th International Conference on Multiple Zeta Values and Related Fields, Changsha, China

Fall 2021–Spring 2022 Tsinghua-BIMSA Learning Seminar on Multiple Zeta Values, Tsinghua University

Other Experiences and Activities

2017 Finisher of Columbia168 ULTRA-TRAIL® THREE GORGES (168 km trail running)

Spring 2013 Exchange student at École Normale Supérieure, Paris, France

Publications and Preprints

14. L. Lai and J. Sprang, *On the $P(t)$-adic Littlewood conjecture in odd characteristics*, arXiv:2606.00633
13. L. Lai, C. Lupu and J. Sprang, *On the irrationality of certain p-adic zeta values*, Res. Math. Sci., **12** (2025), no. 4, Paper No. 77. arXiv:2505.23088
12. L. Lai and J. Li, *A partial result towards the Chowla–Milnor conjecture*, arXiv:2505.12687v2
11. L. Lai, J. Sprang and W. Zudilin, *A note on the irrationality of $\zeta_2(5)$*, arXiv:2505.05005v2
10. L. Lai, *A note on the number of irrational odd zeta values, II*, to appear in J. Théor. Nombres Bordeaux, arXiv:2501.05321v2
9. L. Lai, *Small improvements on the Ball–Rivoal theorem and its p-adic variant*, arXiv:2407.14236v2
8. L. Lai and J. Sprang, *Many p-adic odd zeta values are irrational*, to appear in Michigan Math. J., arXiv:2306.10393v2
7. L. Lai, *On the irrationality of certain 2-adic zeta values*, Int. J. Number Theory, **21** (2025), no. 1, 207–235. arXiv:2304.00816
6. S. Charlton, H. Gangl, L. Lai, C. Xu and J. Zhao, *On two conjectures of Sun concerning Apéry-like series*, Forum Math., **35** (2023), no. 6, 1533–1547. arXiv:2210.14704
5. L. Lai, C. Lupu and D. Orr, *Elementary proofs of Zagier's formula for multiple zeta values and its odd variant*, Proc. Amer. Math. Soc., **154** (2026), no. 01, 11–24. arXiv:2201.09262
4. L. Lai, *On the largest prime divisor of $n!+1$*, Bull. Aust. Math. Soc., **113** (2026), issue 3, 390–403. arXiv:2103.14894
3. L. Lai and L. Zhou, *At least two of $\zeta(5), \zeta(7), \dots, \zeta(35)$ are irrational*, Publ. Math. Debrecen, **101**/3–4 (2022), 353–372. arXiv:2103.00904
2. L. Lai, J.-Y. Li and P. Yu, *On the rigidity of stationary charged black holes: small perturbations of the non-extremal Kerr-Newman family*, J. Differential Geom., **125** (2023), no. 3, 553–612. arXiv:1911.10560
1. L. Lai and P. Yu, *A note on the number of irrational odd zeta values*, Compos. Math., **156** (2020), no. 8, 1699–1717. arXiv:1911.08458